

* Let $F(t)$ be a function of t specified for $t > 0$. Then the Laplace transform of $F(t)$, denoted by $L\{F(t)\}$, is denoted by

$$L\{F(t)\} = f(s) = \int_0^{\infty} e^{-st} F(t) dt \quad (1)$$

where s is a parameter

① If $F(t) = 1$,

$$\text{then } L\{1\} = f(s) = \frac{1}{s}$$

② If $F(t) = t$ then,

$$L\{t\} = f(s) = \frac{1}{s^2}$$

③ If $F(t) = t^n$ then,

$$L\{t^n\} = f(s) = \frac{n!}{s^{n+1}} = \frac{n!}{s^{n+1}}$$

④ If $F(t) = e^{at}$ then,

$$L\{e^{at}\} = f(s) = \frac{1}{s-a}$$

⑤ If $f(t) = e^{-at}$ then,

$$L\{e^{-at}\} = f(s) = \frac{1}{s+a}$$

$$(6) L\{\sin at\} = f(s) = \frac{a}{s^2 + a^2}$$

$$(7) L\{\cos at\} = f(s) = \frac{s}{s^2 + a^2}$$

$$(8) L\{\sinh at\} = f(s) = \frac{a}{s^2 - a^2} \quad s > |a|$$

$$(9) L\{\cosh at\} = f(s) = \frac{s}{s^2 - a^2}$$

$$(10) L\{e^{at} f(t)\} = f(s-a) \rightarrow \text{replace } s \text{ by } s-a$$

Linearity Property

$$L\{C_1 F_1(t) + C_2 F_2(t) + C_3 F_3(t)\} = C_1 f_1(s) + C_2 f_2(s) + C_3 f_3(s)$$

$$* L\{4t^2 - 3\cos 2t + 5e^{-t}\}$$

$$= 4 \cdot \frac{2!}{s^{2+1}} - 3 \cdot \frac{s}{s^2 + 2^2} + 5 \cdot \frac{1}{s+1}$$

$$= 4 \cdot \frac{2}{s^3} - 3 \cdot \frac{s}{s^2 + 4} + \frac{5}{s+1}$$

Ans.

* Find the laplace transform of the following functions?

① e^{at} ② e^{-at}

Solⁿ: $L\{F(t)\} = \int_0^{\infty} e^{-st} \cdot F(t) dt, \quad s > 0$

$$L\{e^{at}\} = \int_0^{\infty} e^{-st} e^{at} dt = \int_0^{\infty} e^{-(s-a)t} dt$$

$$= \left| \frac{e^{-(s-a)t}}{-(s-a)} \right|_0^{\infty}$$

$$= \frac{1}{-(s-a)} (e^{-\infty} - e^0)$$

$$= \frac{1}{-(s-a)} (0 - 1)$$

$$= \frac{1}{s-a}$$

First ~~transform~~ translation or shifting property;

* If $L\{f(t)\} = f(s)$ then, $L\{e^{at} f(t)\} = f(s-a)$

$$\text{Let, } L\{e^{at} f(t)\} = \int_0^{\infty} e^{-st} e^{at} f(t) dt \quad (\text{By definition})$$

$$= \int_0^{\infty} e^{-(s-a)t} f(t) dt \quad \left| \begin{array}{l} \text{Let,} \\ s-a = u \end{array} \right.$$

$$= \int_0^{\infty} e^{-ut} f(t) dt$$

$$= f(u)$$

$$= f(s-a)$$

(proved)

$$* L\{t e^{at}\} = \frac{1}{(s-a)^2}$$

$$* L\{\cos 3t\} = \frac{s}{s^2 + 3^2}$$

$$* L\{e^{-2t} \cos 3t\} = \frac{s+2}{(s+2)^2 + 3^2}$$

$$= \frac{s+2}{s^2 + 4s + 13}$$

$$= \frac{s+2}{s^2 + 4s + 13}$$

$$* \text{ If } L\{F(t)\} = f(s)$$

$$\text{and } g(t) = \begin{cases} F(t-a) & t > a \\ 0 & t < a \end{cases}$$

$$\text{then, } L\{g(t)\} = e^{-as} f(s)$$

* If $L\{F(t)\} = f(s)$, then

$$L\{F(at)\} = \frac{1}{a} f\left(\frac{s}{a}\right)$$

Solⁿ:

$$L\{F(t)\} = \int_0^{\infty} e^{-st} F(t) dt$$

$$L\{F(at)\} = \int_0^{\infty} e^{-st} F(at) dt$$

$$= \int_0^{\infty} \frac{1}{a} e^{-\frac{su}{a}} F(u) du \quad \left| \begin{array}{l} \text{let} \\ at = u \\ t = \frac{u}{a} \\ dt = \frac{1}{a} du \end{array} \right.$$

$$= \frac{1}{a} \int_0^{\infty} e^{-\frac{s}{a} \cdot u} F(u) du$$

$$= \frac{1}{a} f\left(\frac{s}{a}\right)$$

Laplace transform of derivatives:

$$\textcircled{1} L\{F(t)\} = f(s), \text{ then } L\{F'(t)\} = sf(s) - F(0)$$

Solⁿ:

$$L\{F'(t)\} = \int_0^{\infty} e^{-st} F'(t) dt \quad \boxed{\int (u \cdot v)'}$$

$$= \left[e^{-st} F(t) \right]_0^{\infty} + s \int_0^{\infty} e^{-st} F(t) dt$$

$$= 0 - e^0 F(0) + s L\{F(t)\}$$

$$= -F(0) + sf(s)$$

$$= sf(s) - F(0)$$

$$(ii) L\{F''(t)\} = s^2 f(s) - sF'(0) - F''(0)$$

$$(iii) L\{F'''(t)\} = s^3 f(s) - s^2 F'(0) - sF''(0) - F'''(0)$$

$$(iv) L\{F^n(t)\} = s^n f(s) - s^{n-1}F'(0) - s^{n-2}F''(0) - \dots - sF^{n-2}(0) - F^{n-1}(0)$$

$$* L\{F(t)\} = f(s), \text{ then}$$

$$L\{t^n F(t)\} = (-1)^n \frac{d^n}{ds^n} f(s) = (-1)^n f^{(n)}(s)$$

Ex -

$$* \text{Find } L\{t \sin at\}$$

Soln:

$$L\{F(t)\} = f(s)$$

$$\text{Here, } F(t) = \sin at$$

$$L\{\sin at\} = \frac{a}{s^2 + a^2} = f(s)$$

$$L\{t \sin at\} = (-1) \frac{d}{ds} \left(\frac{a}{s^2 + a^2} \right)$$

$$= (-1) \frac{-2as}{(s^2 + a^2)^2} = \frac{2as}{(s^2 + a^2)^2}$$

* Laplace transform of a function division by t

$$\text{If } L\{F(t)\} = f(s), \text{ then } L\left\{\frac{F(t)}{t}\right\} = \int_0^\infty f(u) du$$

Laplace transform of integrals,

$$\text{If } L\{F(t)\} = f(s), \text{ then } \underline{\quad}$$

$$\textcircled{i} L\left\{\int_0^\infty F(u) du\right\} = \frac{f(s)}{s}$$

$$\textcircled{ii} L\left\{\int_0^t \frac{\sin u}{u} du\right\} = \frac{1}{s} \tan^{-1} \frac{1}{s}$$

Inverse Laplace transform:

If $L\{F(t)\} = f(s)$, then $F(t)$ is called an inverse
~~lap~~ laplace transform of $f(s)$, and it is
denoted by $F(t) = L^{-1}\{f(s)\}$

L^{-1} — laplace Inverse operator

$$(i) L^{-1}\left\{\frac{1}{s-a}\right\} = e^{at}$$

$$(ii) L^{-1}\left\{\frac{s}{s^2+a^2}\right\} = \cos at$$

$$(iii) L^{-1}\left\{\frac{1}{s^2}\right\} = \frac{1}{2} t^2 = \frac{1}{2} t^2$$

$$* L^{-1}\left\{\frac{5}{s-3} - \frac{2s}{s^2+25} + \frac{3}{s^2+4} + \frac{1}{s^4}\right\}$$

$$= 5e^{3t} - 2\cos 5t + \frac{3}{2}\sin 2t + \frac{1}{3!} t^3$$

✓ s.s.

$$\star L^{-1} \left\{ \frac{6s-4}{s^2-4s+20} \right\}$$

$$= L^{-1} \left\{ \frac{6(s-2)+8}{(s-2)^2+4^2} \right\}$$

$$= 6 L^{-1} \left\{ \frac{s-2}{(s-2)^2+4^2} \right\} + 2 L^{-1} \left\{ \frac{4}{(s-2)^2+4^2} \right\}$$

$$= 6 e^{2t} \cos 4t + 2 e^t \sin 4t$$

01-12-24

* $Y''(x) - Y'(x) = x$ when $Y(0) = 2$ and $Y'(0) = -3$

Solⁿ:

$$Y''(x) - Y'(x) = x$$

$$\Rightarrow L\{Y''(x) - Y'(x)\} = L\{x\}$$

$$\Rightarrow \{s^2 Y(s) - sY(0) - Y'(0)\} - \{sY(s) - Y(0)\} = \frac{1}{s^2}$$

$$\Rightarrow s^2 Y(s) - sY(0) -$$

$$\Rightarrow s^2 Y(s) - s \cdot 2 + 3 - sY(s) + 2 = \frac{1}{s^2}$$

$$\Rightarrow Y(s) \{s^2 - s\} - 2s + 5 = \frac{1}{s^2}$$

$$\Rightarrow Y(s) \{s^2 - s\} = \frac{1}{s^2} + 2s - 5$$

$$\Rightarrow Y(s) = \frac{1}{s^2(s^2 - s)} + \frac{2s}{s^2 - s} - \frac{5}{s^2 - s}$$

$$= \frac{1}{s^2(s-1)} + \frac{2}{s-1} - \frac{5}{s(s-1)}$$

[partial fraction method]

$$= -\frac{1}{s} - \frac{1}{s^2} - \frac{1}{s^3} + \frac{1}{s-1} + \frac{2}{s-1} + \frac{5}{s} - \frac{5}{s-1}$$

$$Y(s) = \frac{4}{s} - \frac{1}{s^2} - \frac{1}{s^3} - \frac{2}{s-1}$$

$$L^{-1}\{Y(s)\} = L^{-1}\left\{\frac{4}{s} - \frac{1}{s^2} - \frac{1}{s^3} - \frac{2}{s-1}\right\}$$

$$Y(s) = 4 \cdot 1 - 1 \cdot x - \frac{1}{2!} x^2 - 2e^x$$

$$= 4 - x - \frac{1}{2} x^2 - 2e^x$$

Ans.

$$* Y''(x) - Y'(x) = x$$

$$\frac{d^2 y}{dx^2} - \frac{dy}{dx} = x$$

* $Y''(t) + 9Y(t) = \cos 2t$ when,
 $Y(0) = 1, Y(\frac{\pi}{2}) = -1$

Solⁿ: let, $Y'(0) = a$

$$\Rightarrow L\{Y''(t) + 9Y(t)\} = L\{\cos 2t\}$$

$$= (s^2 Y(s) - sY(0) - Y'(0)) + 9Y(s) = \frac{s}{s^2 + 4}$$

$$\Rightarrow s^2 Y(s) - s - a + 9Y(s) = \frac{s}{s^2 + 4}$$

~~$$(s^2 + 9)Y(s) = \frac{s}{s^2 + 4} + s + a$$~~

$$\Rightarrow s^2 Y(s) + 9Y(s) = \frac{s}{s^2 + 4} + s + a$$

$$\Rightarrow Y(s)(s^2 + 9) = \frac{s}{s^2 + 4} + s + a$$

$$\Rightarrow Y(s) = \frac{s}{(s^2 + 9)(s^2 + 4)} + \frac{s}{s^2 + 9} + \frac{a}{s^2 + 9}$$

$$= \frac{1}{5} \frac{s}{s^2 + 4} + \frac{4}{5} \left(\frac{s}{s^2 + 9} \right) + \frac{a}{s^2 + 9}$$

$$\mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\left\{\frac{1}{5} \frac{s}{s^2+4} + \frac{4}{5} \frac{s}{s^2+9} + \frac{a}{s^2+9}\right\}$$

$$\Rightarrow Y(t) = \frac{1}{5} \cos 2t + \frac{4}{5} \cos 3t + \frac{a}{3} \sin 3t$$

when, $Y\left(\frac{\pi}{2}\right) = -1$

$$Y\left(\frac{\pi}{2}\right) = \frac{1}{5} \cos 2 \cdot \frac{\pi}{2} + \frac{4}{5} \cos 3 \cdot \frac{\pi}{2} + \frac{a}{3} \sin 3 \cdot \frac{\pi}{2}$$

$$-1 = \frac{1}{5} \cdot (-1) + \frac{4}{5} \cdot 0 + \frac{a}{3} (-1)$$

$$\Rightarrow -1 = -\frac{1}{5} - \frac{a}{3}$$

$$\Rightarrow \frac{a}{3} = 1 - \frac{1}{5}$$

$$\Rightarrow a = \frac{4}{5} \times 3 = \frac{12}{5} \quad \text{Ans}$$